

AJAY SINGH

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PROFESSIONAL SUMMARY

Highly analytical and results-driven Senior Software Engineer with **6 years of experience** building scalable backend architectures, designing full-stack enterprise applications, and migrating cloud workloads (AWS, Azure, GCP). Adept at accelerating the Software Development Life Cycle (SDLC) by integrating modern AI technologies, including Machine Learning, Deep Learning, GitHub Copilot, and Cursor AI. Proven track record of architecting distributed microservices, improving system performance by up to 40%, delivering enterprise-grade features for global Fortune 500 clients, and leading technical initiatives through robust High-Level Design (HLD) and Low-Level Design (LLD).

TECHNICAL SKILLS

Programming Languages:	C#, C++, Python, Go, JavaScript, TypeScript, Julia, C
AI, ML & Modern Tools:	Machine Learning, Deep Learning, Generative AI (LLMs), GitHub Copilot, Cursor AI, Wit.ai, API.ai
Web & Frameworks:	.NET, .NET Core, ASP.NET MVC, Angular, HTML5, CSS3, Bootstrap
Databases & Cloud:	SQL Server, MySQL, Redis, Elasticsearch, Azure, AWS, GCP
Core Competencies:	System Design (HLD & LLD), Design Patterns, System Architecture, Microservices, Data Structures & Algorithms, Object-Oriented Programming (OOP), CI/CD, Unit Testing
Tools & Version Control:	Docker, Git, TFS, Bitbucket, JIRA, SonarQube, Visual Studio, Selenium, Kibana

PROFESSIONAL EXPERIENCE

Granicus	<i>11/2024 - Present</i>
Software Engineer III	
- Spearheaded the System Design (HLD & LLD) and development of three complex, highly scalable features from scratch for the Vision product in the Gov-Access division, enhancing public sector data accessibility for millions of users. - Accelerated development lifecycles by 30% by extensively utilizing AI-assisted development tools like GitHub Copilot and Cursor for rapid prototyping, intelligent code refactoring, and comprehensive unit test generation. - Designed and integrated Python microservices to run advanced data extraction and machine learning routines alongside the primary C#/.NET Core platform, improving backend processing speeds by 25%. - Mentored and groomed 3 junior engineers on complex system architecture principles, scalable database design, and the implementation of robust Design Patterns for modern software delivery.	
Xplor Technologies	<i>Pune, IN 12/2023 - 11/2024</i>
Software Development Engineer II	
Drove end-to-end SDLC sprints within the Core Engineering team for Xplor Recreation, encompassing requirement gathering, architectural POC creation (HLD), robust API development, and seamless AWS server deployments.	

- Proactively diagnosed and resolved 20+ critical production bottlenecks across distributed environments, leading to a 35% improvement in overall system performance and significantly reducing system downtime.
- Injected **Machine Learning (ML)** models to refine user personalization in recreation booking, heavily utilizing Python for backend data analysis while maintaining the robust core application via C#/.NET.
- Enhanced code coverage to over 90% by incorporating AI-driven test frameworks via **Cursor** and native unit testing modules, ensuring zero regression failures during major enterprise release cycles.

Icertis

Pune, IN | 07/2020 - 12/2023

Software Engineer

- Engineered and deployed 15+ complex enterprise features from scratch for the OM-App product in the Core ICI & Application Engineering Team, serving 50+ global Fortune 500 customers independently.
- Architected scalable RESTful APIs and distributed background services utilizing optimal **Design Patterns** in C# and .NET Core, optimizing unstructured data retrieval times by 40% using Redis and Elasticsearch.
- Modernized the UI/UX utilizing Angular and TypeScript while integrating advanced predictive contract analytics and automated compliance tracking logic.
- Automated over 100 complex use cases via Python scripts and Selenium, reducing manual regression testing efforts by 50% and boosting CI/CD deployment frequency.

ACADEMIC PROJECTS

Corruption Free Nation (Citizen Engagement Platform)

- Developed a high-impact web portal aimed at democratizing municipality decision-making in India, utilizing HTML5/CSS, JavaScript, and Bootstrap for user-centric interfaces.
- Integrated intelligent conversational chat-bots using Wit.ai, API.ai, and **Machine Learning** models written in Julia and Python to interact dynamically with users and analyze local issue trends.

Mitigating Black Hole Attacks in WSN

- Engineered an ADIOS-based Node-Resident Expert System using NS2 and TCL on Ubuntu to securely detect and mitigate Black Hole network intrusions.
- Achieved a 95% detection accuracy for sensor network vulnerabilities, establishing a reliable security baseline.

EDUCATION

Army Institute of Technology (University of Pune)

Pune, IN | 06/2016 - 06/2020

Bachelor of Engineering (B.E) in Information Technology — **CGPA: 8.97/10**

Relevant Coursework: Data Structures & Algorithms, OOPS, OS, Data Science & ML, Cloud Computing, System Design, DBMS

D.P.S JJN (CBSE)

Rajasthan, IN | 04/2014 - 05/2015

Senior Secondary School Education — **85.60%**

ACHIEVEMENTS & ADDITIONAL INFORMATION

- Passionate about algorithmic problem solving; successfully solved **500+ coding problems** on Codeforces, GeeksforGeeks, and LeetCode.
- Consistently leverage modern generative AI workflows and continuous learning to stay at the forefront of backend system architecture and machine learning engineering.